

Scope Fit Audit

Paper: “Institutional Observability Under Scaled AI Governance” **Auditor:** Technology in Society handling editor perspective **Date:** June 4, 2026

Criterion 1: Topic Fit

Assessment: [GREEN] GREEN

The paper addresses AI governance and institutional oversight — a core topic in Technology in Society’s recent research directions. The journal explicitly lists “AI ethics and AI governance” and “algorithmic accountability and fairness” among its active areas.

Evidence: - Journal publishes on AI governance frameworks, algorithmic accountability, and compliance infrastructure - Recent issues feature papers on AI ethics, governance-by-design, and audit mechanisms - The paper’s focus on institutional observability under scaled deployment directly engages the technology-society-governance intersection

Criterion 2: Methodological Fit

Assessment: [YELLOW] YELLOW

The paper uses state-space modeling, queueing theory, and Monte Carlo simulation. These are rigorous methods but are more commonly seen in CS/engineering venues than social science journals. Technology in Society accepts theoretical work, but the mathematical density may challenge some readers.

Evidence: - Journal accepts “scholarly and theoretical approaches” per aims and scope - However, most published papers use qualitative, survey-based, or mixed methods - The paper’s mathematical formalism is unusual for this venue but not disqualifying

Mitigation: Explanatory prose surrounds equations. The limitations section is honest about the theoretical nature.

Criterion 3: Audience Fit

Assessment: [YELLOW] YELLOW

The paper targets external auditors, regulatory bodies, governance boards, and AI system vendors. This aligns with Technology in Society’s audience of “decision-makers” and policy actors. However, the technical density (O(N) notation, Kingman approximations, KDE plots) may alienate some readers.

Evidence: - Journal aims to “enable decision-makers to engage thoughtfully with technologies” - The paper names specific policy actors and actionable interventions - The SEDI index is designed for use by non-technical auditors

Mitigation: The cover letter and highlights emphasize policy implications over technical details.

Criterion 4: Novelty Fit

Assessment: [GREEN] GREEN

Oversight saturation as a structural failure mode of AI governance has not been formalized this way before. The SEDI index is a genuinely novel contribution — a computable governance observability metric from public records.

Evidence: - No existing paper formalizes the throughput asymmetry between AI decision generation and human review capacity - The 2.71x saturation multiplier (from Kingman + log-normal service) is a new result for governance contexts - SEDI operationalizes governance observability as a load-dependent variable — a novel framing

Criterion 5: Journal Fit (Overall)

Assessment: [GREEN] GREEN — FIT

The paper sits at the technology-society-governance intersection that Technology in Society cultivates. It is not a pure CS paper (the contribution is governance, not algorithms), not a pure STS paper (it includes formal modeling), and not a pure policy paper (it derives principles from mathematical analysis).

Comparison to alternative venues: - **NeurIPS/AAAI:** Too governance-focused, not enough algorithmic novelty - **Social Studies of Science:** Too mathematical, not enough ethnographic/qualitative - **Research Policy:** Too technical, not enough empirical policy analysis - **Technology in Society:** Right interdisciplinary fit — technology mechanism + governance implications + policy actors

Verdict

SUBMIT — The paper fits Technology in Society's scope. The primary risk is the mathematical density, which is mitigated by explanatory prose and honest limitations. The novelty and policy relevance are strong.